

EDAV Assessment 2

1. Exploratory Data Analysis

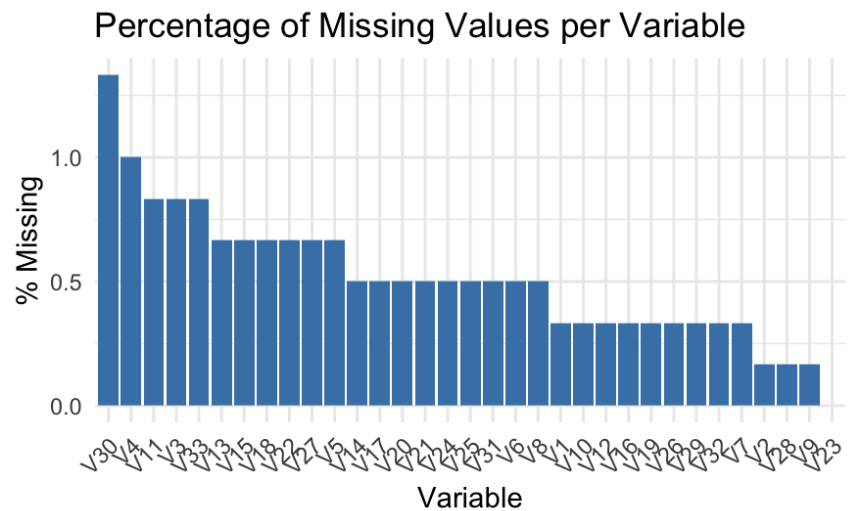
1.1 Dataset Overview

The sampled dataset comprises 600 observations across 33 numeric variables. Variables V1 - V28 have ranges spanning hundreds (e.g. V1: -189 to 332), while V29 - V33 are confined to single-digit intervals (± 5). The max-to-min raw variance ratio is 2250.6, so standardisation is needed.

1.2 Missingness

Overall missingness is 0.5%. No variable exceeds 1.33% missing (V30). The `vis_miss()` map and UpSet plot confirm that missingness is scattered with negligible co-missingness patterns. Over 500 of 600 observations are fully complete, and the maximum per-observation missingness is 2 variables.

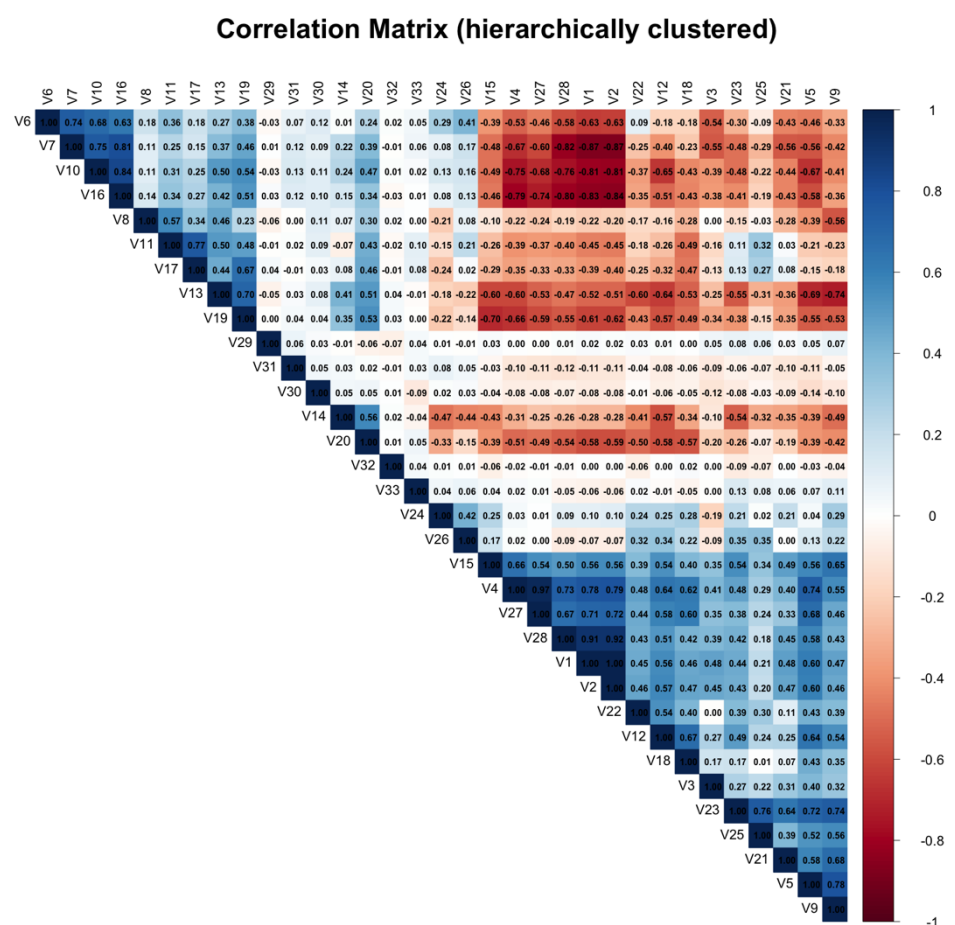
Figure 1: Percentage of missing values per variable.



We deduce missingness appears Missing Completely at Random (MCAR). No imputation is necessary for PCA.

1.3 Correlation Structure

The hierarchically-clustered correlation matrix reveals two dominant blocks of positively correlated variables, with strong negative correlations at their intersection. This opposing signal is consistent with chromosome arm morphology, where these features may represent two arms which contribute complementary measurements. There is a block with near 0 correlated variables, which may be candidates for removal. Figure 2: Correlation matrix (hierarchically clustered). Two variable blocks visible.



1.4 Principal Components Analysis

PC1 captures 37.7% of variance, confirming the dominant correlation structure. An elbow appears after PC3. Reaching 80% cumulative variance requires 10 components, meaning variance may come from more than the two correlated groups.

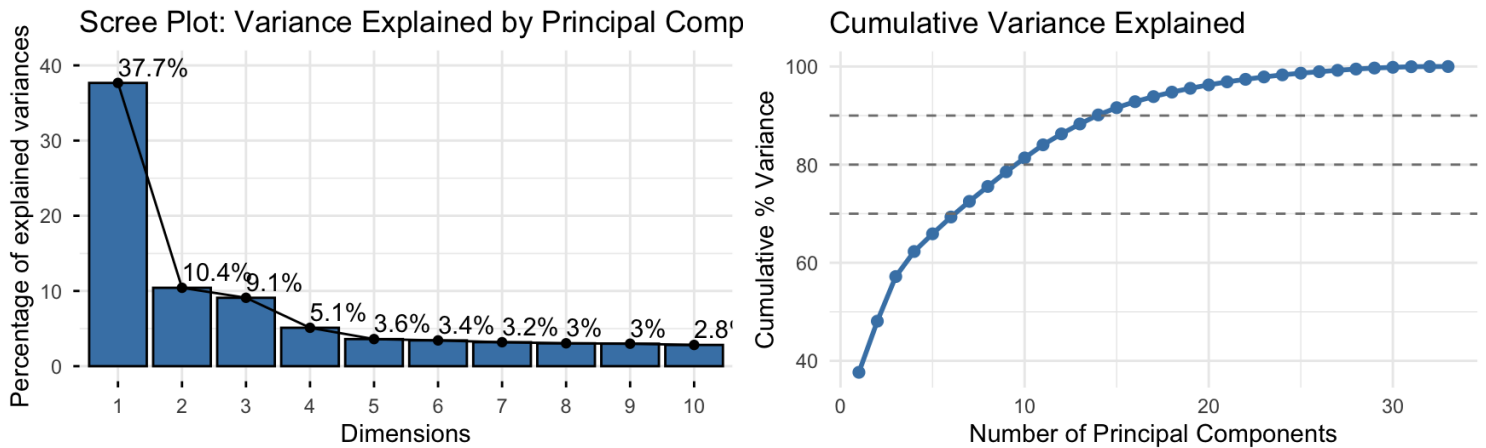


Figure 3: Scree plot (left) and cumulative variance explained (right).

The PC1, PC2 score plot shows a continuous circular pattern rather than discrete clusters. Observations with high $\cos^2$ sit at PC1 extremes; those with low $\cos^2$ cluster centrally, suggesting their variation is captured by later components. PCA 3, 4 indicates mild variation, so further variation might be available in further PCs.

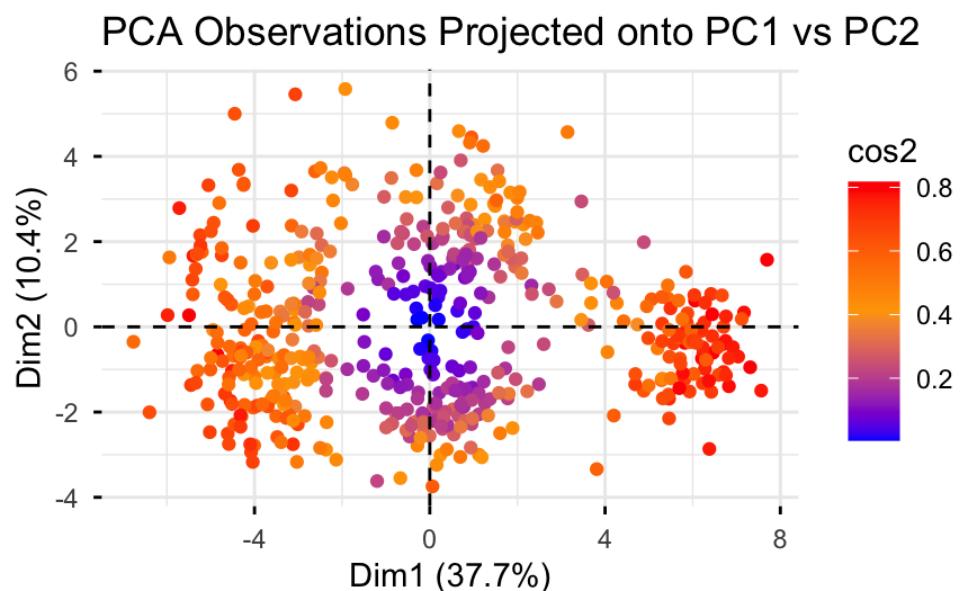


Figure 4: PCA score plot coloured by $\cos^2$.

1.5 Variable Removal & Outliers

V29 and V31 are flagged by both near-zero variance (<1% of median) and low PCA loadings (<0.15 across the first 5 PCs). These are removed. We choose to retain V30, V32, V33 for conservativeness, as they may carry banding-pattern information. Modified Z-score analysis ($|M| > 3.5$) finds only isolated outliers consistent with biological heavy-tailed distributions; no observations are removed.

1.6 Scaling Decision

With a max/min variance ratio of 2250.6, all variables are standardised to zero mean and unit variance before clustering.

2. Cluster Structure and Comparison

2.1 Standard Euclidean Route

Missing-data strategy: Column-mean imputation. Missingness is scattered (approximately MCAR), so mean imputation introduces minor distortion. Casewise deletion would reduce n from 600 to 508: a 15% loss would be more damaging to k-means centroid estimation than the modest variance underestimation from imputation.

Figure 5: Elbow plot for the Standard Euclidean route ($k = 2-7$).

The elbow is not sharp, but $k = 4$ provides the last practically meaningful improvement (10.8%), after which marginal gains fall below 7%. The PC1, PC2 projection also visually suggests four groups.

Selected $k = 4$

Figure 6: Euclidean clusters ($k = 4$) with convex hulls in PCA space.

2.2 Observed-Only Distance Route

The renormalised Euclidean distance is computed on standardised (un-imputed) data, with the scaling factor $p/|O_{ij}|$ correcting for varying numbers of shared observed coordinates. Classical MDS embeds the distance matrix into 2D (47.7% variance explained). k-means is run on the MDS coordinates.

Selected $k = 4$ as the curve flattens noticeably from $k = 4$ to $k = 5$. (Not shown here)

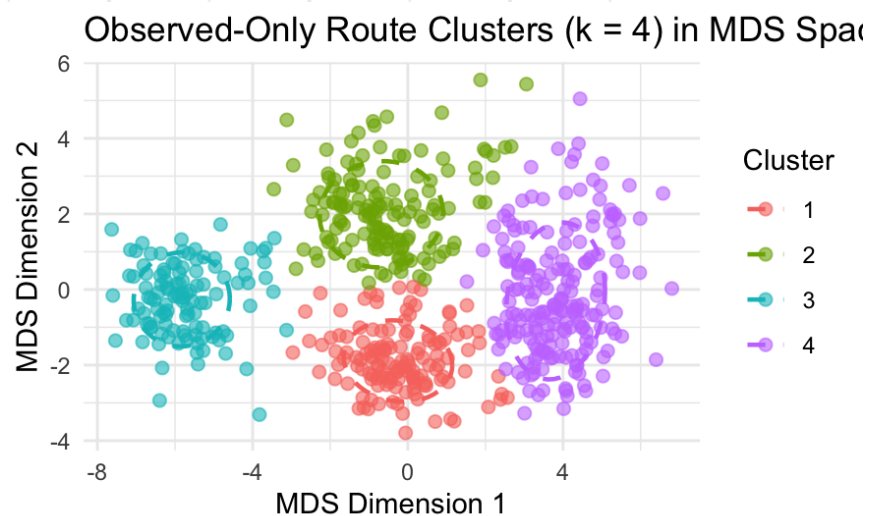
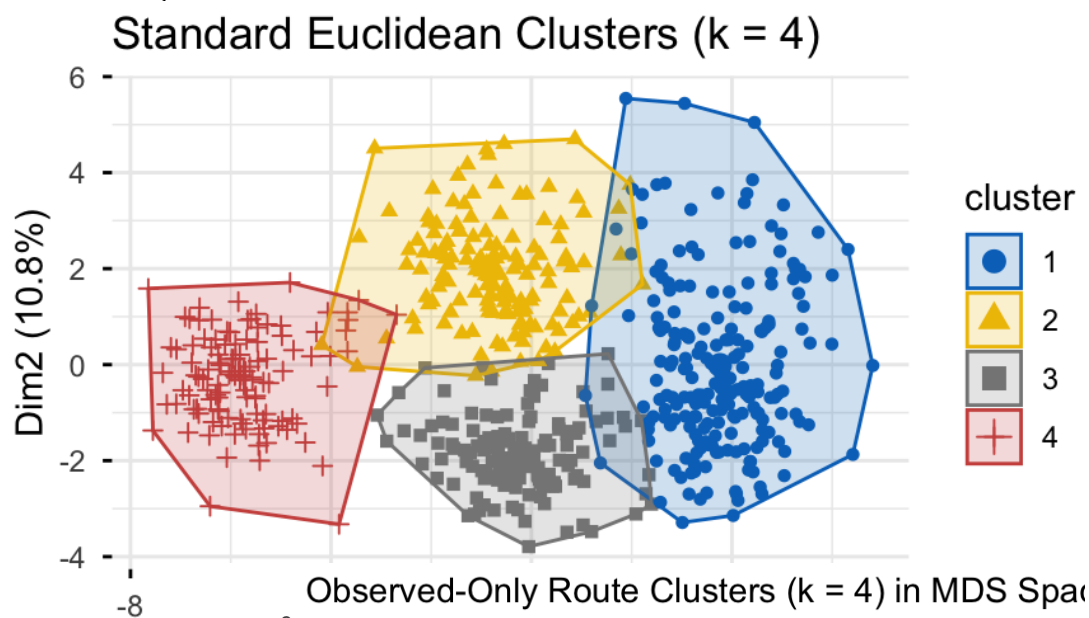
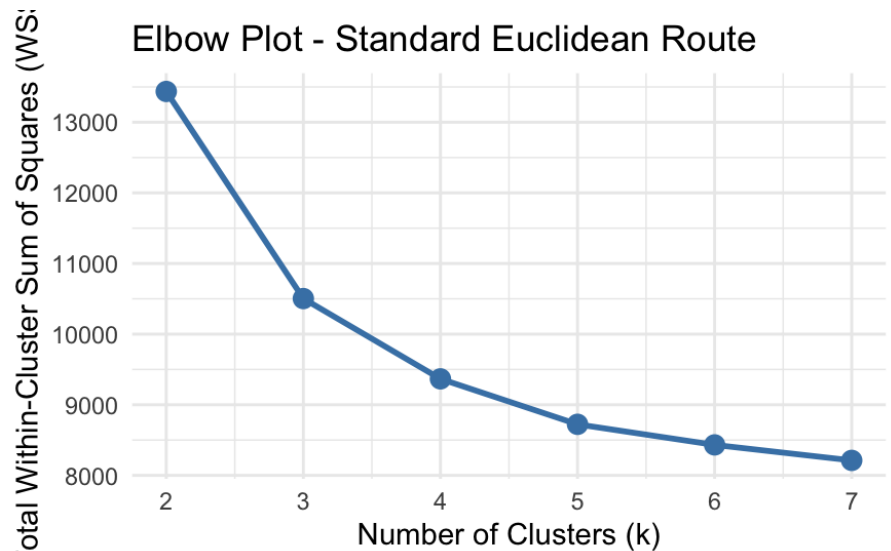


Figure 7: Observed-Only clusters ($k = 4$) in MDS space.

2.3 Comparison

Our cross-tabulation of the two cluster vectors shows strong diagonal agreement, indicating not many differing assignments. We calculate this to be 4.0%.

	MDS 1	MDS 2	MDS 3	MDS 4
Euc 1	197	10	3	0
Euc 2	1	136	4	2
Euc 3	2	0	131	1
Euc 4	0	1	0	112

Table 3: Cross-tabulation of cluster assignments.

Differing assignments: 24/600 (4.0%). Both routes independently select $k = 4$ and assign 96% of observations identically.

This strong agreement indicates that the cluster structure is stable and seems to reflect genuine biological groupings, not artefacts from preprocessing.

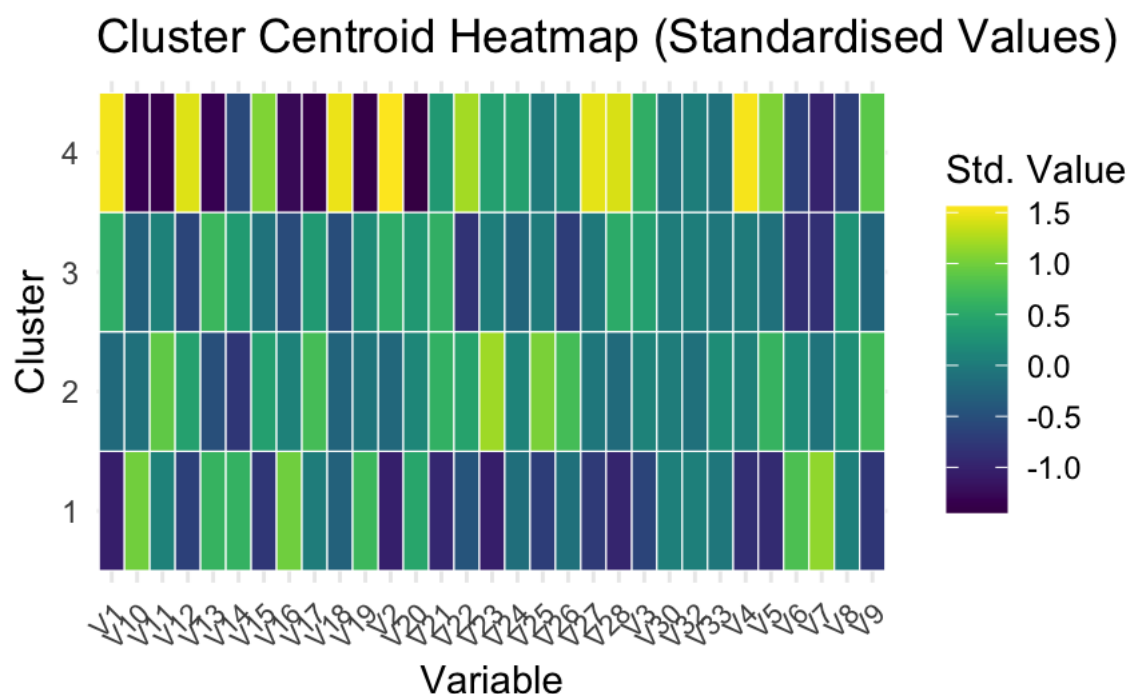
Mild differences arise from: (i) mean imputation artificially tightening Euclidean clusters by strengthening correlations; (ii) PCA maximising variance vs MDS preserving distances, leading to slightly different cluster boundaries.

3. Cluster Interpretation

3.1 Cluster Prototypes

The Euclidean pipeline ($k = 4$) is selected as the preferred solution, as outcome is similar between the methods, and Euclidean is an easier and less complex method.

Figure 8: Cluster centroid heatmap (standardised values).



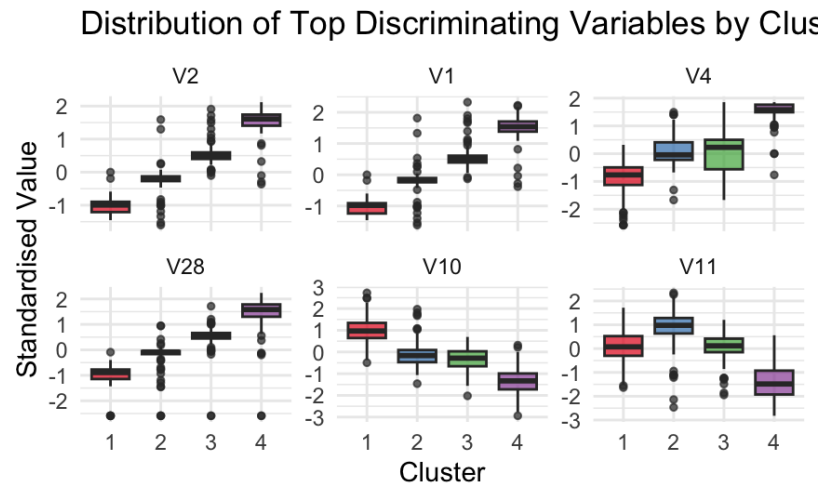
Cluster 4 shows the most distinctive profile, with extreme values across most variables. Clusters 1 and 4 appear to be opposites, while Clusters 2 and 3 occupy intermediate positions. This could indicate 4 distinct features

leading to the profiles, or a continuous single feature grouping. For the sake of our analysis, we continue with the 4 groupings approach, stating that they represent 4 distinct chromosome classes.

3.2 Discriminating Variables

Figure 9: Distribution of top discriminating variables (V2, V1, V4, V28, V10, V11) by cluster.

- **Cluster 1:** Consistently low on V1, V2, V4, V28; high on V10. Most discriminating and cohesive structure indicated in previous analysis
- **Cluster 2:** Near-zero on most variables; notably high V11. Could represent a specific mildly discriminating property.
- **Cluster 3:** Moderate positive on V1, V2, V4, V28; negative on V10, V11. Opposing profile to Cluster 1.
- **Cluster 4:** Highest values on V1, V2, V4, V28; most negative on V10, V11. Represents an extreme opposite of Cluster 1.



3.3 Within-Cluster Variability

Cluster 1 has the highest raw Within SS, but this is driven by its larger size ($n = 210$). After normalising, all four clusters show comparable average spread ($\sim 14.6 - 16.8$ per observation), i.e. all four clusters are comparatively tight and have cohesion.

Table 4: Within-cluster sum of squares and cluster sizes.

Cluster	Within SS	Size	Avg Within SS
1	3,526	210	16.79
2	2,146	143	15.01
3	1,962	134	14.64
4	1,732	113	15.33

3.4 Conclusion: Number of Classes

Finding: 4 chromosome classes are present in this dataset.

- Both independent pipelines (Euclidean and MDS) converge on $k = 4$, with 96% assignment agreement.
- Each cluster exhibits a distinct centroid profile across the key discriminating variables.
- The elbow plot shows diminishing returns beyond $k = 4$ (marginal improvement drops from 10.8% to 6.9%).
- Normalised within-cluster variability is comparable across all four groups.

3.5 Uncertainty

The 2D PCA projection compresses 33 dimensions, reaching 80% variance requires 10 components. Clusters that overlap in 2D may be well-separated in higher dimensions. The gradient structure observed across clusters could suggest these are not four discrete classes but rather points along a continuous grouping that k-means has discretised.